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ECOV2 Chlorophyll Fluorometer Characterization Sheet

Channel No.: 1
Date: 1/8/2026 S/N: ECOV2-10116
Wavelength: 470 nm Firmware: 3.10.1

Chlorophyll concentration expressed in µg l^{-1} can be derived using the equation:

CHL (µg chlorophyll l^{-1}) = Scale Factor * (Raw Output - Dark counts)

Dark counts, this characterization					**54**	counts
Dark counts Range		min:	**53**	max:	**55**	counts
Scale Factor (SF)					**4.794E-03**	µg CHL l^{-1} $count^{-1}$
Resolution	Raw output < 5000	**1.15**	counts		**5.51E-03**	µg CHL l^{-1}
	Raw output ≥ 5000	**1.00**	counts		**5.03E-02**	µg CHL l^{-1}
Standard Error as % of Scale Factor					**0.13**	%
Specified Range:				min:	**400**	µg CHL l^{-1}
Maximum Output (see note below)					**3296**	µg chlorophyll l^{-1}
Scale Ratio					**10.49**	

- **Raw Output**: Signal output of the instrument. The outputs of the high and low gain stages have been appropiately offset and scaled before combining as follows: If Raw Ouput < 50000, Raw Output = counts[HI_GAIN]. If Raw Output ≥ 50000, Raw Output = (counts[LO_GAIN]-DarkCounts[LO_GAIN])*ScaleRatio + DarkCounts[HI_GAIN] Note that DarkCounts llsted on this sheet are DarkCounts[HI_GAIN]

- **Dark Counts**: The average of 1 minute of collected counts[HI_GAIN] data after 15 minutes of warm-up obtained by covering the detector with the black tape and submersing the sensor in a grounded water bath. The characterization dark counts value is not expected to be the same as the dark counts after sensor integration. Use the dark counts measured after sensor integration for data processing.

- **Dark Counts Range**: Minimum and maximum values recorded in the data collected to determine the dark count value.

- **Scale Factor**: The slope of the linear regression of the target addition series comparing this instrument to a reference instrument.

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- **Resolution:** Standard deviation in counts of the data collected to determine the dark counts presented as counts and engineering units by muliplying by the scale factor. For standard deviations of less than one count, resolution is the same as the scale factor.
If Raw Output ≥ 50000, then resolution should be scaled by Scale Ratio (Resolution * ScaleRatio)

- **Standard Error as % of Slope:** 100* (Standard error of slope from linear regression) / (Scale Factor) . This value represents uncertainty of this calibration relative to that of the reference sensor. It does not represent total uncertainty in the measurement.
- **Specified Range:** Calibrated range of operation
- **Maximum Output**: Maximum potential scaled output of the instrument. This value is generally greater than the specified range.

- **ScaleRatio:** Amplifcation ratio of high:low gain channels = (counts[HI_GAIN]-DarkCounts[HI_GAIN]) / (counts[LO_GAIN]-DarkCounts[HI_GAIN]) . When converting Raw Output to engineering output, this value is not needed; low gain raw data has already been appropriately offset and scaled.

Characterization Methodology:

We perform a multipoint linear regression analysis of the output of this instrument to the output of a calibrated reference instrument using an additive sequence of increasing loadings of uranine dye. The goodness of fit is determined by the standard error which must be less than 1% of the scale factor. The reference units were calibrated to extracted chlorophyll by measuring the fluorescence response in a monoculture of Thalassiosira weissflogii in controlled light conditions and in exponential phase growth. The chlorophyll concentration from the cultures was determined by the extraction and fluorescence method. This establishes an instrument response to chlorophyll concentration scale factor for the reference unit, with respect to the particular culture used. Caution should be observed in using the calibrated value without reference to grab samples.

Note on Maximum Output:

The maximum number of counts that the instrument can output is approximately 160,000. In most natural waters the potential range of the instrument is greater than the measurable range due to interference factors, e.g. the absorbance length scale. Caution should be used for data sets in which the reported values exceed the specified range. Grab samples should be used to confirm values above the specified range.

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